

Aditya Shirsatrao

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Professional Summary

B.Tech Artificial Intelligence and Data Science undergraduate (CGPA 8.5/10) with 6+ production-grade full-stack projects, 366+ solved LeetCode problems, and hands-on experience building scalable backend and ML systems. Currently studying Japanese (JLPT N5, preparing for N4) with strong interest in the Japanese tech ecosystem. Seeking a Software Engineering / Backend / AI internship in Japan to contribute to high-scale, product-focused development.

Technical Skills

Languages: Python, Java, C++, JavaScript (ES6+), TypeScript, SQL

Backend: FastAPI, Node.js, Express.js, REST APIs, WebSockets, Redis, Kafka, Celery, Message Queues

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS

AI/ML: PyTorch, TensorFlow, HuggingFace Transformers, scikit-learn, OpenCV, RAG, LangChain, LLMs

Databases: PostgreSQL, MySQL, MongoDB, ChromaDB

DevOps: Docker, Kubernetes, GitHub Actions CI/CD, AWS (EC2/S3/Lambda), Linux

Concepts: DSA, OOP, System Design, Microservices, Distributed Systems

Projects

SentinelX – AI Cybersecurity Platform

GitHub

FastAPI, HuggingFace Transformers, Whisper, Celery, Redis, PostgreSQL

May 2026

- Built a production-grade multi-channel threat-detection backend (email/SMS/voice) using a multi-model ML pipeline: DistilBERT for NLP classification, OpenAI Whisper for speech-to-text, and LLM enrichment
- Engineered a weighted 4-component risk-scoring engine with async Celery + Redis task queues; won **1st Place** at Hack to Future 3.0 (600+ teams, INR 50,000 prize)

Tracelify – Cross-Platform Error Tracking & Observability

GitHub

Python, FastAPI, React, AWS Bedrock, PostgreSQL

Apr 2026

- Designed and published a multi-language error-tracking SDK (PyPI: tracelify-sdk v1.0.0) with native support for Python, Java, Kotlin, Go, C++, JavaScript
- Built a FastAPI backend with multi-tenant RBAC, Redis caching, and LLM-powered automated error reports; **1st Runner-Up** at Orchathon 2K26 (800+ teams)

Fraud Detection System

GitHub

FastAPI, Autoencoders, React, D3.js

Dec 2024 – Present

- Architected a real-time anomaly-detection engine (unsupervised autoencoders) on 500k+ transactions; reduced inference latency to **<50ms** at 100 requests/second

Privacy-First LLM Assistant (RAG)

RAG, Llama 3, ChromaDB, Ollama

GitHub

Nov 2024 – Dec 2024

- Built a local Retrieval-Augmented Generation system with Hybrid Search + cross-encoder re-ranking, improving retrieval precision by **40%** on technical corpora

DevGuard – RBAC API Gateway

Express.js, JWT, PostgreSQL, Prisma

Code

Oct 2024 – Nov 2024

- Built a secure backend API gateway with role-based access control and multi-session token logic for enterprise applications

Experience

Fullstack Web Development Intern

Jul 2025 – Jan 2026

Unified Mentor Pvt. Ltd

Remote

- Architected an AI-powered code review platform with Kafka queues and Redis caching, reducing analysis latency by **45%**
- Engineered a real-time collaborative IDE (WebSockets, Socket.IO, CRDT) with sub-100ms synchronization for concurrent users
- Built a distributed ML inference platform with autoscaling endpoints and monitoring, improving throughput by **60%**

AI/ML Lead

Sep 2025 – Present

Google Developer Groups on Campus (GDG-NKOCET)

Solapur, India

- Lead AI/ML workshops and mentorship for a **200+ student** community

Education

B.Tech in Artificial Intelligence and Data Science Technology

Expected Graduation: 2027

N. K. Orchid College of Engineering &

CGPA: 8.5/10

- Relevant coursework: Data Structures & Algorithms, Databases, Operating Systems, Distributed Systems, Machine Learning, Neural Networks, OOP

Languages

- **Japanese:** JLPT N5 (currently preparing for N4)
- **English:** Professional working proficiency
- **Hindi / Marathi:** Native

Achievements & Certifications

- IEEE First-Author Publication presented at ICCTWC 2026; 3 national hackathon wins (600–800+ teams)
- 50+ open source PRs merged across Google (OSS-Fuzz), Microsoft (VibeVoice), Apache (DataFusion), Hedera Hashgraph (Hiero SDK)
- NPTEL Programming in Java (Elite+Gold, 91/100); JPMorgan Chase & Electronic Arts SWE Virtual Experience; Kaggle AI Agents Intensive Course

Relocation / Visa

- Willing to relocate to Tokyo / anywhere in Japan for an internship
- Eligible for a Designated Activities (internship) visa; committed to reaching JLPT N3+ while working